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運用商業新聞與文字探勘技術預測公司永續治理表現

Predicting ESG Performance of Firms Using Business News:

A Text Mining Approach

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Business News: A Text Mining Approach

本論文係林楷翊君（學號 R08725026）在國立臺灣大學資訊管理學系、所完成之碩士學位論文，於民國 110 年 7 月 30 日承下列考試委員審查通過及口試及格，特此證明

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誌謝

在碩士的這兩年真的過得很快，從碩一開始對論文的懵懂，到碩二慢慢學習如何發現問題、思考解方、到最後的解決問題，回想起來很像是一趟學習的旅程，更感謝每一個願意給我幫助與學習機會的人。首先最要感謝指導教授—魏志平教授，在每一個禮拜的報告中總是不厭其煩地和我們討論當期論文，並且去理解每一位同學各式各樣新奇的想法，並以最專業謹慎的態度來分析解答我們的每一個問題。除了學業之外，也謝謝教授在工作、生活等各層面上提供很多想法與價值觀與我們一起討論，更以身作則鼓勵我們多運動保持健康。同時也感謝虹鈞一直以來的幫忙與管理，讓我們能夠無後顧之憂的專心準備論文。

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林楷翊 謹識



Abstract

ESG is commonly agreed to be one of the most transformative issues today. Not only the environmentalists but also the consumers, investors, and even the enterprises start to pay attention to the ESG performance of a firm and its supply chain companies. With the increased awareness of ESG, various ESG rating agencies have been established. However, the process that an ESG agency rates a firm is very time-consuming due to the huge amount of efforts required in preparing ESG reports and other related information. With the advances of natural language processing (NLP) methods, we attempt to propose an automatic model that can classify the ESG performance of firms using only business news or other publicly available information. We experiment with different methods including vector space models and deep learning-based methods. Besides, given the power of the pre-trained language model, we also investigate BERT and ESG-BERT, which are pre-trained on the general corpus and further pre-trained on the ESG investing text corpus respectively. Our developed methods, data collection, and evaluation results are detailed.

Keywords: ESG; ESG performance classification; text mining; vector space model; deep learning; natural language processing; pre-trained language model



摘要

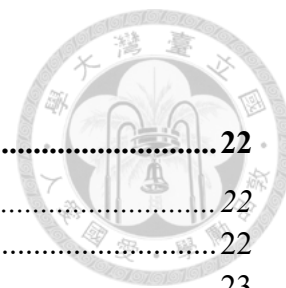
環境社會治理（ESG）原則已經儼然變成是當今最具變革性的議題之一。無論是環保主義者、消費者、投資人或甚至是企業都開始關注一家公司與其供應鏈的永續治理表現。隨著 ESG 越來越受到關注，有些企業組織開始建立起 ESG 的評分機構。然而，在現行的 ESG 機構評分架構下，由於需要收集公司的 ESG 報告和其他相關資訊，評分機構需要花費大量的時間與勞力資源才能夠完成評分。因此，隨著自然語言處理（NLP）的快速發展，我們希望能夠利用 NLP 提出一個利用新聞文件或其他公開可取得的資訊，自動分類一家公司的 ESG 表現的技術。明確而言，我們嘗試使用向量空間（Vector Space Model）與基於深度學習(Deep Learning-Based Model)的模型去建立這套評分系統。除此之外，有鑒於預訓練語言模型應用在各項不同語言任務的成功，我們也研究了 BERT 跟其延伸模型 ESG-BERT。在此篇論文中，我們將詳細的描述了我們採用的文字探勘模型、資料集合以及評估結果。

關鍵字：環境社會治理；企業 ESG 表現分類預測技術；向量空間模型；深度學習；自然語言處理；預訓練語言模型



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Chapter 1

Introduction



1.1 Background

ESG, namely Environmental, Social, and Governance, has become one of the standards to evaluate a company's behavior, strategy, or business model and is used to determine its future financial performance by socially conscious investors.^{1,2} "Environmental" describes how a company manages energy usage and other environmental impacts toward their products. "Social" is used to evaluate how a company fosters its employees, suppliers, and customers, etc. "Governance" is a criterion on how a company deals with its policy, including internal control, decision making, board-level gender diversity, and shareholders' rights. The growing trend could be shown in the emergence of the sustainability reports, many of them are now highlighting the importance of not only "Social Responsibility" but the way how they govern their business.

¹ <https://sphaera.com/spark/the-importance-of-esg-strategy/>

² <https://www.investopedia.com/terms/e/environmental-social-and-governance-esg-criteria.asp>

Firms' ESG performance can affect their corporate financial performance (CFP). For example, Friede et al. (2015) collected and analyzed 2,200 individual studies and showed that about 90% of studies suggested a non-negative ESG-CFP relation. Capelle-Blancard & Petit (2019) and Serafeim & Yoon (2021) both find a positive market reaction to positive ESG news and events, a negative reaction to negative ones.

According to J.P. Morgan³, the broadly defined ESG market is expected to reach \$45 trillion; by Bloomberg⁴, ESG assets may hit \$53 trillion by 2025, a third of global Assets under Management. Due to the emerging attention, more and more investors start to notice the importance of ESG and take ESG performance into their investing strategies as well, such as Cathay MSCI Taiwan ESG Sustainability High Dividend Yield ETF (0078)⁵, which choose companies with outstanding ESG performances as their investment portfolio targets.

1.2 Research Motivation

With the increased awareness of ESG, various ESG rating agencies have been established, such as MSCI ESG Ratings, S&P Global Green Evaluation⁶, and Truvlaue ESG Ranks. For instance, in MSCI ESG Ratings, each company is given an ESG rank within {AAA, AA, A, BBB, BB, B, CCC}. They adopt a rule-based methodology to evaluate one firm's

³ https://www.jpmorgan.com/insights/research/covid-19-esg-investing?source=cib_pr_cc_esginvestinga0620

⁴ <https://www.bloomberg.com/professional/blog/esg-assets-may-hit-53-trillion-by-2025-a-third-of-global-aum/>

⁵ <https://www.cathaysite.com.tw/funds/etfglobal/indexinfo.aspx?fc=CN>

⁶ <https://www.spglobal.com/ratings/en/index>

ESG performance based on different key issues. We take the soft drink industry as an example and demonstrate part of the MSCI's rating standards in Figure 1.

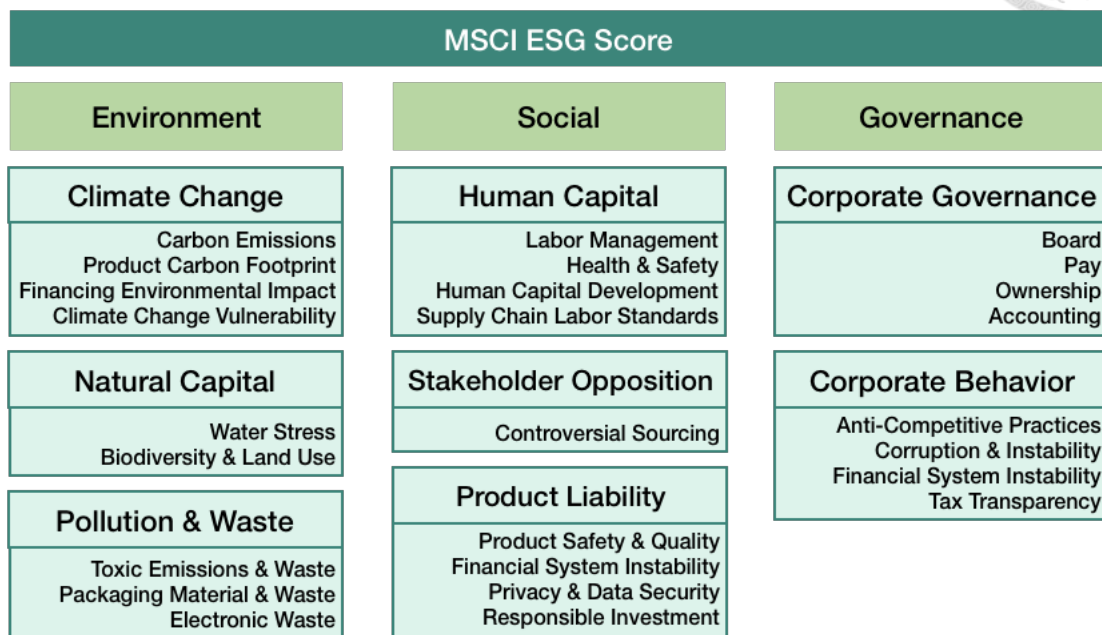


Figure 1: MSCI ESG Rating Model and Underlying Key Risk Issues for the Soft Drinks Sub Industry (Part.) ⁷

The current practice proceeds as follows: First companies prepare their annual ESG reports and related questionnaire surveys and then submit these documents to a specific ESG rating agent. Subsequently, the rating agent relies on experienced analysts to rate the ESG performance of each company by scrutinizing the submitted report and the information reported in the external sources (e.g., news).

⁷ <https://www.msci.com/our-solutions/esg-investing/esg-ratings>



Due to the huge amount of effort required in the preparation of ESG reports, only a few very-large-sized companies can afford to participate in ESG ratings. For those companies who are not able to dedicate that much effort to ESG rating processes, it is hard for agencies and investors to evaluate their ESG performance. Furthermore, because, in the current practice, the ESG rating is evaluated once to twice a year by ESG rating agencies, it is difficult to monitor companies' ESG performances “dynamically” by catching up with updated events and news. Accordingly, in this research, we attempt to propose and develop an ESG rating technique that can classify the ESG performance of a company using only business news or other publicly available information. Besides, we believe our proposed technique can also provide those rating agencies with another ESG insight by keeping up with any updated ESG related news about the company.

1.3 Research Objective

In this research, we aim to propose an automatic technique that can classify the ESG performance of a company only using publicly available information (specifically, news reports in this study). With the advances of natural language processing (NLP), we will develop two approaches for our research task (i.e., classifying a company's ESG performance). First of all, we will experiment with the approach of vector space model in conjunction with various supervised learning algorithms (e.g., k -nearest neighbors, random forest, XGBoost, and backpropagation neural network). Second, we will explore the deep

learning-based approach to see if we can build more effective classification models. We will further compare the classification effectiveness of the two proposed approaches.



Chapter 2

Literature Review



2.1 Text Mining on ESG domain

It is not until recently that companies and investors started looking into ESG investment, so there have not been many studies on predicting the ESG performance of companies. Liew et al. (2014) used the traditional text mining technique (specifically, TF-IDF) to identify sustainability trends and practices by analyzing firms' sustainability reports. They assumed that the frequent use of sustainability-related terms in the report reflected the company's concern and emphasis on sustainability issues. Shahi et al. (2014) proposed the idea of CSR report scoring by counting how many GRI⁸ standards a company discloses in the report via a document classification mechanism.

Kiriū & Nozaki (2020) built a word structuring model to evaluate CSR reports and compared the word counts to ESG performance rated by Thomson Reuters⁹ via regression. However, they indicated that by merely reviewing the CSR report to rate the ESG performance would have the potential problem of greenwashing (Delmas & Burbano, 2011).

⁸ <https://www.globalreporting.org/>

⁹ <https://www.thomsonreuters.com/en.html>

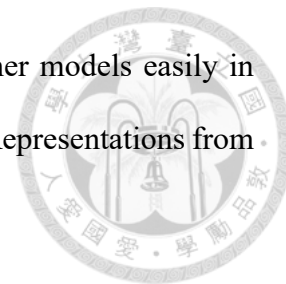
Greenwashing refers to the situation that companies are not vigorously engaged in environment protection but provide a false impression to the society as they are environmentally friendly firms. To diminish the hidden worries of greenwashing, we will take business news as our data for constructing an ESG classification model which is a fairer source in contrast to ESG reports prepared by companies.

Nugent et al. (2020) developed an ESG topic classification model (i.e., identifying the ESG topics discussed in an article) using pre-trained language model BERT (Devlin et al., 2018) and data augmentation approaches. They further pre-train a BERT model on financial and business articles with a 715M word corpus from Reuters News Archive. The model can detect and classify ESG topics with an 84% accuracy when an ESG related article is given. Following the study of Nugent et al. (2020), we will not only use the vector space approach but also explore the deep learning-based approach (e.g., BERT) and evaluate their classification effectiveness.

2.2 Pre-trained Language Model

The principle behind a pre-trained language model is to construct a model which can understand a focal domain and be applied to various NLP downstream tasks, such as classification, question answering, named entity recognition tasks, etc. The language model is first fed with large unlabeled data using some self-supervised tasks as the training target(s). Then the model is fine-tuned with respect to the data pertaining to a downstream task. Because the language model is already trained once in a large corpus, it can catch the

semantic meaning from the dataset more easily and outperform all other models easily in some tasks. In the thesis, we will explore BERT (Bidirectional Encoder Representations from Transformers) as one of our deep learning-based models.



2.2.1 BERT (Bidirectional Encoder Representations from Transformers)

BERT has its debut in 2018, achieving the state-of-the-art performance in many tasks and brings the pre-trained language model for downstream tasks to another level. In general, BERT is pre-trained on a general corpus (e.g., Wikipedia or book corpus). BERT uses the transformer mechanism and defines two language modeling tasks, including Masked Language Modeling (MLM) and Next Sentence Prediction (NSP). MLM is the task to predict randomly masked tokens in a sequence rather than predicting the next one. NSP is the task that the model receives pairs of sentences and predicts whether the second sentence is the next sentence of the first one in the original document.

Taking the advantage of the combination of attention model, MLM, and NSP task, BERT forces the transformer encoder to deeply understand the language and assigns the representation for each token more precisely. The question answering example is shown in Figure 2 to illustrate how BERT is implemented and down-streamed to the target task.

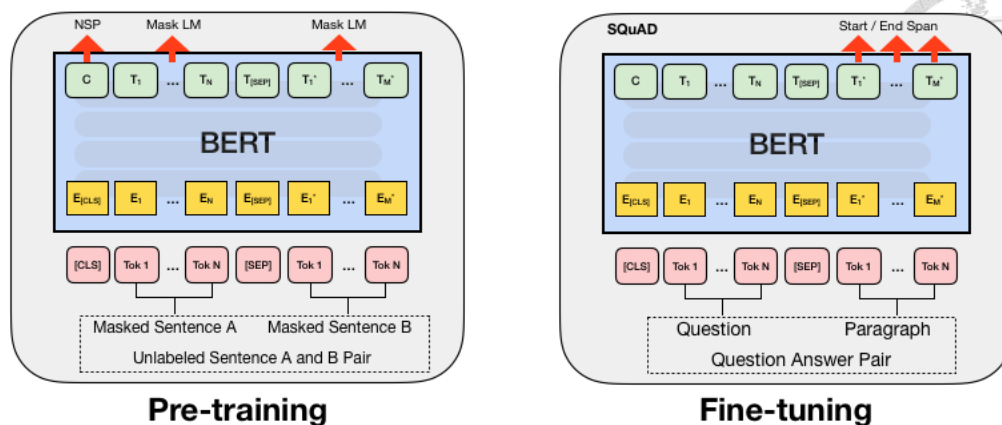
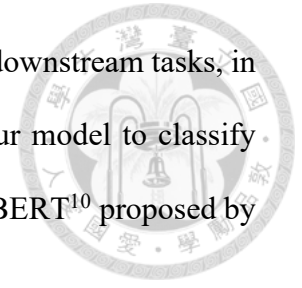


Figure 2: *Pre-training and Fine-Tuning Procedures for BERT (Devlin et al., 2018)*

2.2.2 BERT on Specific Domains

Sun et al. (2019) conduct a series of experiments to investigate fine-tuning strategies for different downstream tasks. They first use BERT and further pre-train the model with MLM and NSP tasks on a domain-specific corpus. After further pre-training, they propose multiple fine-tuning strategies and achieved state-of-the-art performance in eight classification datasets. Aside from further pre-training and fine-tuning the model, there are also other experiments adopting the architecture of BERT and pre-training the model on a domain-specific corpus from the scratch. For example, Beltagy et al (2019) introduced SciBERT which is initialized with the BERT architecture and directly pre-trains on a scientific dataset; Lee et al. (2020) introduce BioBERT, which is pre-trained on biomedical corpora. Both of the models outperform the original BERT in various domain-specific downstream tasks.

As BERT has reached the state-of-the-art performance in many downstream tasks, in this research we will also conduct an experiment utilizing BERT in our model to classify companies' ESG performance. In addition, we will also adopt the ESG-BERT¹⁰ proposed by Mukherjee, which is further pre-trained on ESG investing text corpus.



¹⁰ <https://towardsdatascience.com/nlp-meets-sustainable-investing-d0542b3c264b>

Chapter 3

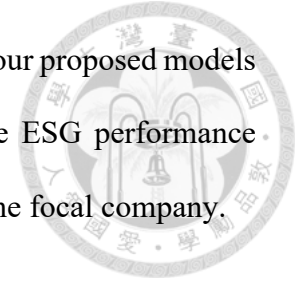
Methodology



As there is not enough data for us to do such ranking tasks as agencies, we regard our task as a classification task. Further details will be discussed in Section 4.1.3. Besides, there does not exist many related research on rating a company's ESG performance using the text mining approach, so we propose various models to process the ESG news and evaluate which one will better predict the ESG rating: Vector Space Model, News Embedding via BERT Encoder Model, BERT for Classification and LSTM with Embedding Layer. Aside from using the news vectors or representations as our main features, we also take the sector to which a company belong into considerations. As there are 11 different company sectors in our dataset, we turn the variable of company sector into a vector via one-hot encoding and append the vector of company sectors to the processed news vectors/representations as the overall features for constructing an ESG classification model.

In addition, ESG ratings are rated every single year in general. Thus, we partition the ESG news of a company into different segments based on the timestamps at which the company was rated. For instance, if a company A has been rated in 2016-10 and 2017-09, then we collect the ESG news from 2016-11 to 2017-09 as the company-year data for 2017.

In the following sections, we will describe the detailed design of our proposed models for constructing a classification model that can be used to predict the ESG performance category of a company on the basis of ESG-related news pertaining to the focal company.



3.1 Vector Space Model

The architecture of vector space model is shown in Figure 3. We first concatenate all ESG news associated with a specific company-year into a news collection as the company news data of the year. After acquiring multiple news collections, we take them as our input data at the company-year level.

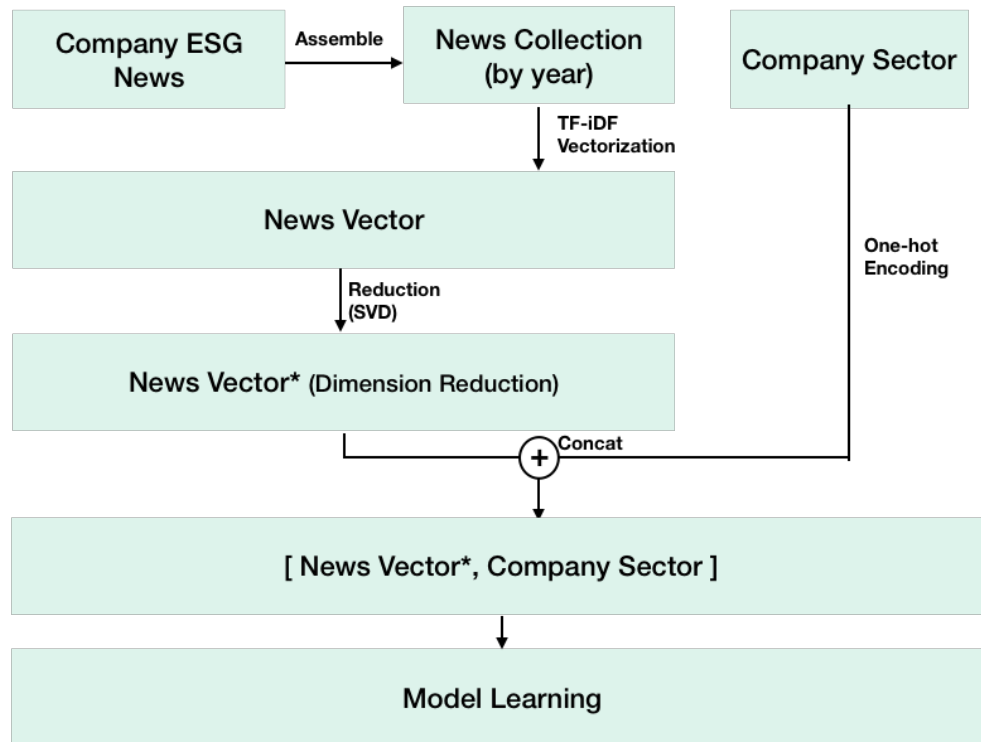


Figure 3: *The Architecture of Vector Space Model*

Subsequently, we adopt the mechanism of bag of words and TF-IDF, which can turn the news collections into vectors. First, we remove stop words from the news collections and apply 1-gram calculation to build a dictionary of the term frequencies and document frequencies. Accordingly, we use the dictionary to performance TF-IDF calculation. The calculation process is shown in Equation 1.

$$tfidf_{i,j} = tf_{i,j} \times idf_i \quad (1)$$

$$tf_{i,j} = \frac{n_{i,j}}{\sum_k n_{k,j}}$$

$$idf = \log \frac{|D|}{|j: t_j \in d_j|}$$

where tf = term frequency, idf = inverse document frequency

t_i = term i , $n_{i,j}$ = the number of term i in document j , D = document set

As the dimension of the news collection is very sparse (i.e., the number of terms is greater than 200k in our dataset), we then apply a dimension reduction mechanism to help filter noise features and preserve the main features from the original vector. Specifically, we employ Singular Value Decomposition (SVD) for dimension reduction, which is derived from the idea of eigenvalue and eigenvector calculation. SVD calculation process is shown in Equation (2).

$$X_{ESG(m \times n)} = U_{ESG(m \times m)} \Sigma_{ESG(m \times n)} V_{ESG(n \times n)}^T \quad (2)$$

where U_{ESG} and V_{ESG}^T denote the left-singular and right-singular vectors respectively. Σ_{ESG} is the matrix of singular values. Since X_{ESG} 's dimension n could be large, Equation (2) can be approximated as Equation (2.1) using a small value k for n .

$$X_{ESG(m \times n)} \approx U_{ESG(m \times k)} \Sigma_{ESG(k \times k)} V_{ESG(k \times n)}^T \quad (2.1)$$

We just need to make the production of $U_{ESG(m \times k)} \Sigma_{ESG(k \times k)} V_{ESG(k \times n)}^T$ similar to $X_{ESG(m \times n)}$ by selecting our ideal dimension k instead of restoring the original $X_{ESG(m \times n)}$. In this way, we can efficiently reduce the dimension of vectors while still keeping most of the important features from the original one. Accordingly, in this research, we feed news collection vectors into SVD to transform them into dimension reduced vectors and take them as our input to a supervised learning algorithm (e.g., k -nearest neighbors, random forest, XGBoost, and backpropagation neural network) to build a classification model.

3.2 News Embedding via BERT Encoder Model

As we have mentioned in Section 2, many prior research focusing on various downstream NLP tasks adopt BERT as a pre-trained language model and further fine-tune the model to the task-specific dataset, reaching the state-of-the-art performance. In BERT,

each input (i.e., ESG news) will first be tokenized into word pieces and fed into the BERT tokenizer of the custom dictionary to get *Token Embeddings*, *Segment Embeddings*, and *Position Embeddings*. Given the embeddings, the encoder can generate the representation for each token, including [CLS] and [SEP]. We will use the [CLS]'s output from the encoder as the representation of the ESG news in this method.

The model architecture of this BERT-based method for ESG performance classification is shown in Figure 4. In this proposed method, we first adopt the BERT-base-model (uncased) as our pre-trained language model. We feed each news article into the BERT tokenizer and encoder to get the [CLS] representation for the news article without fine-tuning the BERT model. However, BERT's input has a maximum length limit of 512 tokens, [CLS] and [SEP] included. It means that we need to decide which 510 tokens to be selected as our BERT input tokens, if the length of an input news article is too long. Because some of the news data have a brief commercial advertisement of news media in the beginning, we will abandon the first 20% content of the news article and select the first 510 tokens from the remaining content of the article. The equation is thus assumed in Equation (3).

$$X_{ESG} = ESGNews[StartPos: StatPos + 510] \quad (3)$$

$$\text{where } StartPos = \begin{cases} 0 & \text{if } l \leq 510 \\ l - 510 & \text{if } 510 < l \leq 640, l = \text{length of } ESGNews \\ l \times 0.2 & \text{if } 640 > l \end{cases}$$

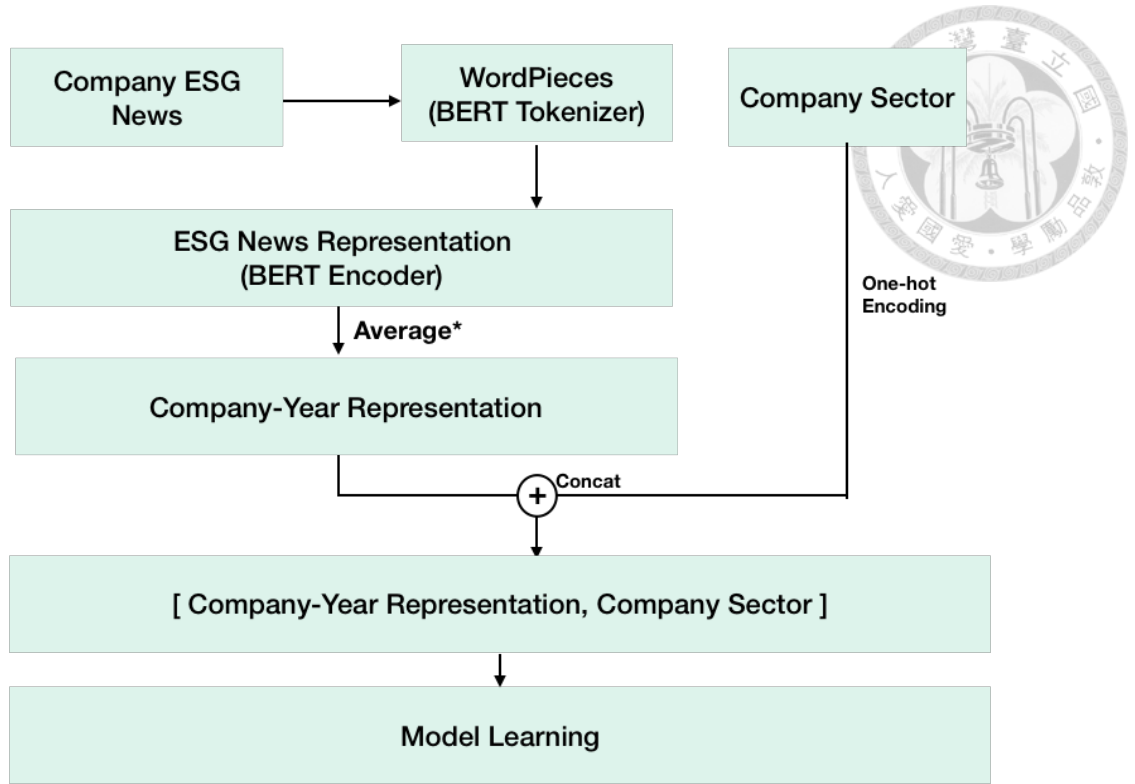


Figure 4: *The Architecture of News Embedding via BERT Encoder Model*

After obtaining the representation vector for each news article for a specific company-year, we propose two schemes to aggregate these representation vectors into an aggregate representation for the focal company-year (i.e., company-year representation). The two schemes include normal average and ESG weighted average.

3.2.1 Normal Average

In the normal average scheme, we simply take the average of all representation vectors, each of which is for a news article pertaining to the focal company-year. The formula for this scheme is shown in Equation 4.

$$CYrep_i = \frac{\sum_{j=1}^N Newsrep_{i,j}}{N}, \quad (4)$$

where $CYrep$ is the aggregate representation vector for the focal company-year, i is the i th element in the representation vector, $Newsrep_{*,j}$ is the representation vector of the news article j , and N is the number of the news articles of the focal company-year.

3.2.2 ESG Weighted Average

In the ESG weighted average scheme, we calculate each news' ESG attentions using the unsupervised method. First, we acquire three ESG related documents collected from Wikipedia (i.e., referred to as E-description, S-description, and G-description documents). These three documents are attached in Appendix A, Appendix B, and Appendix C respectively. We then feed these documents into BERT encoder respectively and obtain three representations for the environmental, social, and governance aspects, respectively. Second, we compute the cosine similarity score between the representation of a news article and the representations for the E-description, S-description, and G-description documents, respectively. We then select the highest similarity score as the news' weight for the weighted average calculation, which also indicates the relevance between the news' topic and ESG. The equation is listed in Equation 5.

$$CYrep_i = \frac{\sum_{j=1}^N Newsrep_{i,j} \times w_j}{\sum_{j=1}^N w_j}, \quad (5)$$

where $w_j = \max(SE_j, SS_j, SG_j)$, $SE_j = \text{similarity}(\text{Newsrep}_j, E\text{-Rep})$, Newsrep_j is the representation vector of the news article j , $SS_j = \text{similarity}(\text{Newsrep}_{i,j}, S\text{-Rep})$, $SG_j = \text{similarity}(\text{Newsrep}_{i,j}, G\text{-Rep})$, and $E\text{-Rep}$, $S\text{-Rep}$, and $G\text{-Rep}$ denote the representation vectors for the E-description, S-description, and G-description documents, respectively.

Aside from using the general BERT, we also adopt ESG-BERT in our proposed method, which has been further pre-trained on unstructured sustainability text corpora. Because of the difference in the corpora for pre-training, we believe ESG-BERT may achieve a greater classification effectiveness than the general BERT does. The corpora for pre-training are shown in Table 1.

Table 1: *Pre-training Corpora for BERT and ESG-BERT*

Model	Corpora
BERT	Wikipedia + Book Corpus
ESG-BERT	BERT + ESG Investing Text Corpus

3.3 BERT for Classification

Different from the method mentioned previously, we now adopt the whole BERT architecture and append a fully connected layer on top of BERT to make predictions. The model architecture is shown in Figure 5.

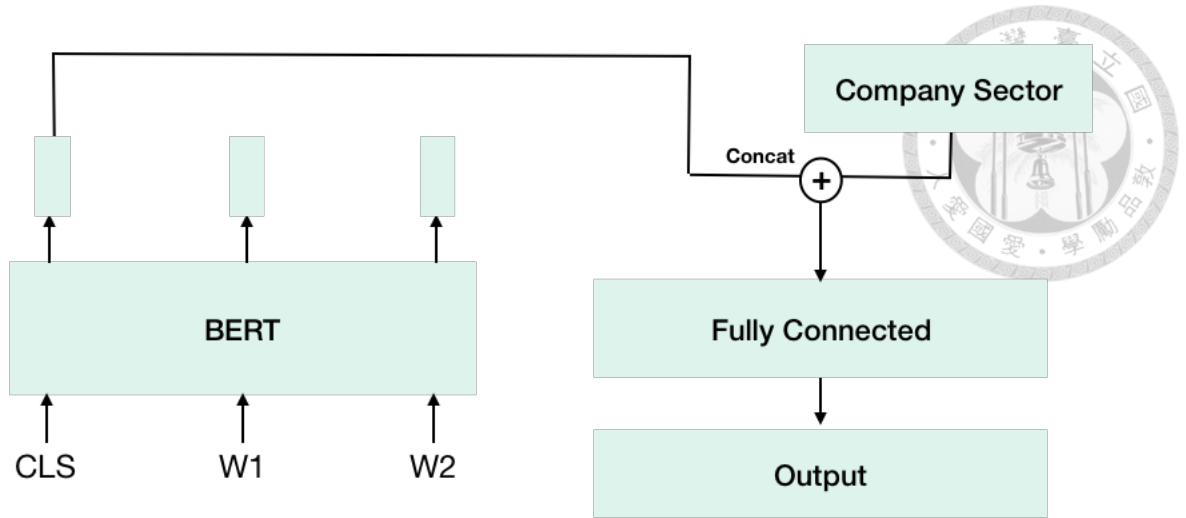


Figure 5: *The architecture of BERT for Classification*

We also face the same constraint of BERT's input length limitation, which we will use Equation 3 to solve the problem. However, there exists another constraint for our task in BERT model architecture: basically, every data sample (x) needs to have an output (y) for model training. For our task, multiple news articles from one company-year will be fed into BERT, but only one company-year ESG rating category will be the output for model training (i.e., $(x_1, x_2, x_3, \dots, x_N) \rightarrow y$), which does not match the architecture of BERT.

Therefore, we set the batch size of a constant as large as we could. We then randomly select a constant number of ESG news articles from the same company-year as our data and feed them into the same batch. Also, we assume that every news article is associated with the output class as the company-year's same ESG rating category (i.e., $(x_1, x_2, x_3, \dots, x_N) \rightarrow (y_1, y_2, y_3, \dots, y_N) = (y, y, y, \dots, y)$).

The modified BERT for classification model is shown in Figure 6 to illustrate our training process. Since news articles from the same company-year will be fed into the same batch for training, the model's weights can be updated for each company-year together.

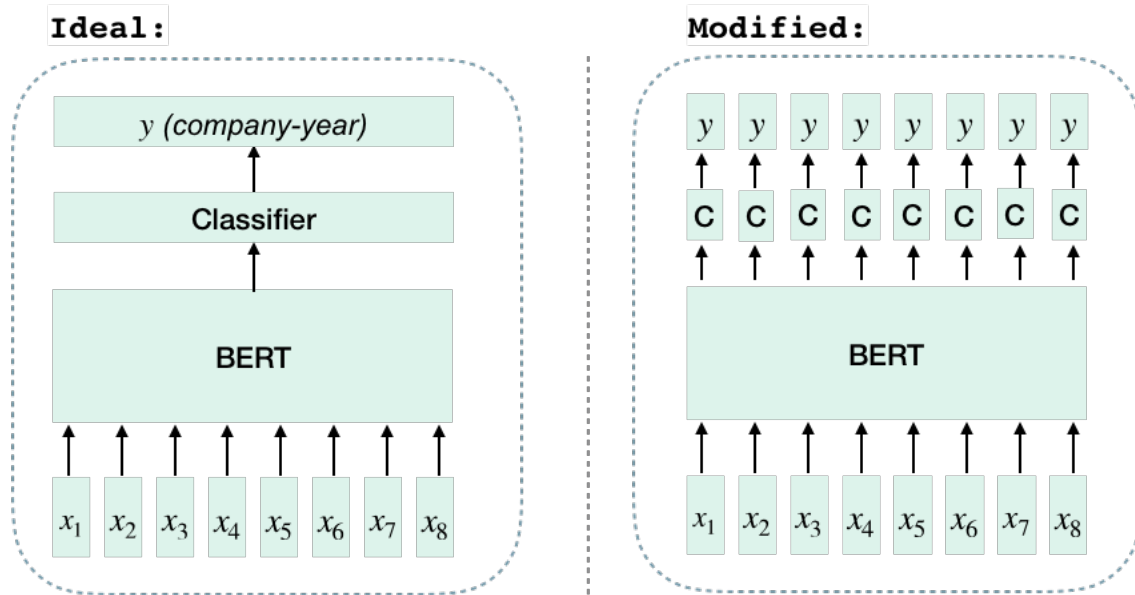


Figure 6: *Modified Model of BERT for Classification*

3.4 LSTM with Embedding Layer

In this proposed method, we apply a deep learning-based mechanism with embedding layer, bidirectional LSTM layers, and fully connected layers. The model architecture is shown in Figure 7. First, we use an embedding layer to encode each word in a news article by indices. Since embedding layers can only encode and learn the word representations, we then feed the word embeddings for the words in a news article into a bidirectional LSTM to learn the representation for the respective news article.

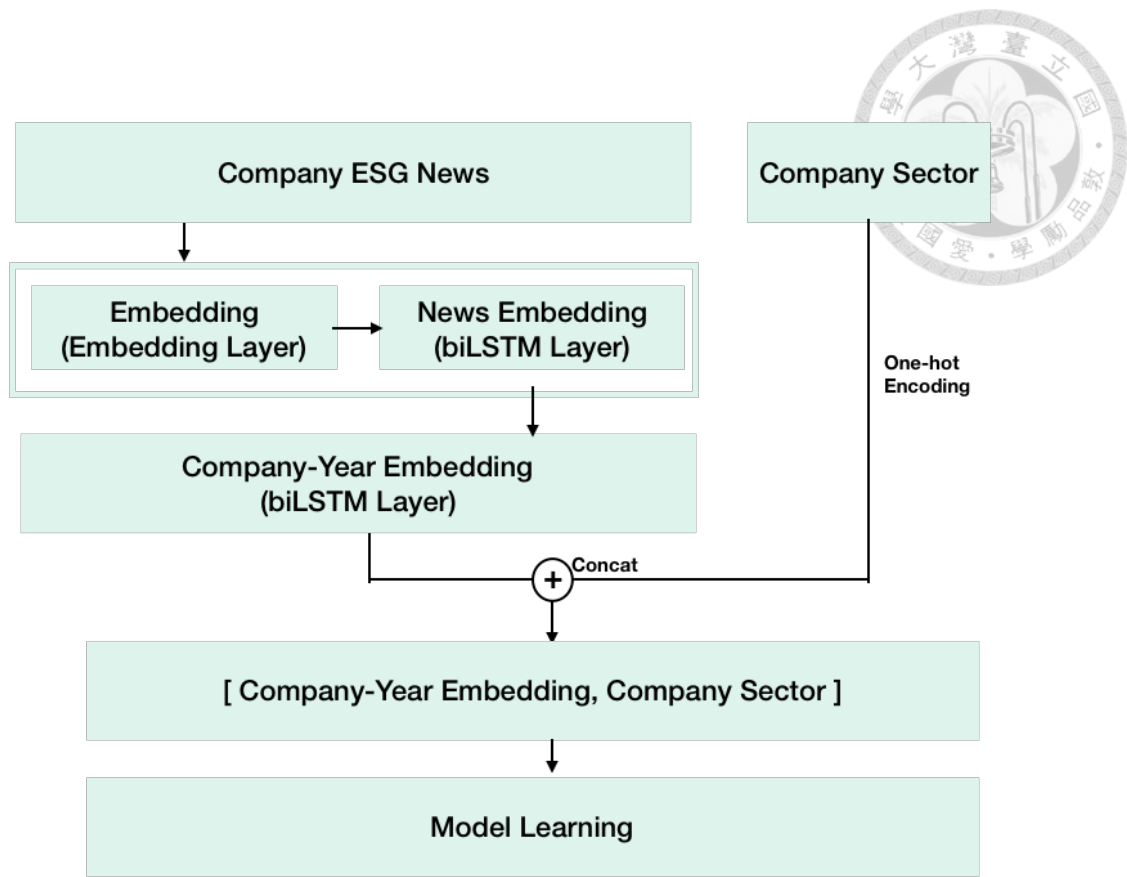


Figure 7: *The Architecture of LSTM with Embedding Layer Model*

After getting multiple news representations for a focal company-year, we feed them into another bidirectional LSTM layer to learn the representation of the specific company-year. On top of the bidirectional LSTM layer, we append two fully connected layers for our classification task. We then backpropagate the model's loss to the two fully connected layers, the bidirectional LSTM layers, and the embedding layer to update the model's weights.

Chapter 4



Empirical Evaluation

4.1 Data Collection

In this section, we will describe how we collect our data and conduct statistical analysis of our dataset.

4.1.1 Company Selection

In choosing companies, we aim to select large-sized companies to ensure we can acquire enough news articles and ESG performance scores of the companies. Therefore, we target the companies listed in S&P 500 Index¹¹ and Forbes Global 2000¹². The S&P 500 Index is a market-capitalization-weighted index of the 500 largest public-traded companies in the U.S. On the other hand, Forbes Global 2000 has measured the world's 2000 largest public companies in terms of assets, market value, sales, and profits. Finally, we select companies from S&P 500 in 2018 and Forbes Global 2000 in 2020.

¹¹ <https://www.investopedia.com/terms/s/sp500.asp>

¹² <https://www.forbes.com/lists/global2000/>



4.1.2 Company News Data

ProQuest is a library-like platform that provides 20 million pages of global, national, regional, and specialty newspapers¹³. We use the ProQuest news database to acquire business news articles related to ESG from 2016 to 2020, whose sources include New York Times, Wall Street Journal, The Economist, Sloan Management Review, The Washington Post, etc. We select the news data whose subject containing ESG related issues (i.e., climate change, product carbon footprint), which is provided by ProQuest. Afterwards, since there are a lot of news talking about stock performance of one company, we filtered the stock market news by a rule-based approach.

4.1.3 Company ESG Scores

We acquire the company's ESG performance rating for the year of 2016 to 2020 from MSCI ESG Ratings. As Section 1.2 mentioned, each company-year is assigned a rating among {AAA, AA, A, BBB, BB, B, CCC} to indicate the company's ESG performance in the specific year. We remove those companies that do not have any news in our news sources or do not have MSCI ESG Ratings in 2016~2020. Afterward, only 588 companies remain and a dataset with 2050 company-years (e.g., Amazon in 2019) and 36,152 news articles is created.

¹³ <https://www.proquest.com/>

Some companies have few ESG-related news articles in one year (i.e., news data for one company-year), so we would like to remove those company-years with a small number of news articles. The statistical analysis of the number of company-years and the amount of news articles with different thresholds of news amount filter are shown in Figure 8 and Figure 9 respectively.

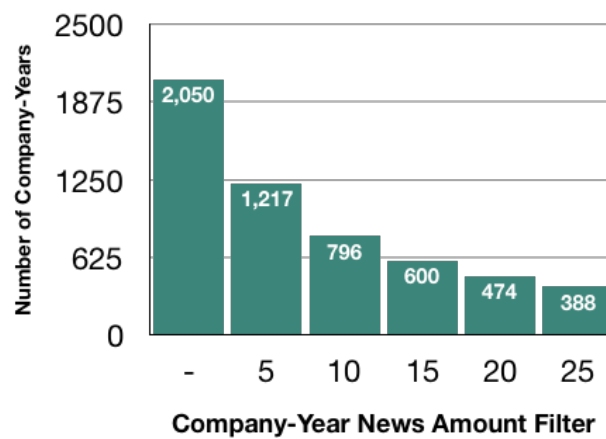


Figure 8: *Statistical Summary of Number of Company-years with Different Thresholds of Company-year News Amount Filter*

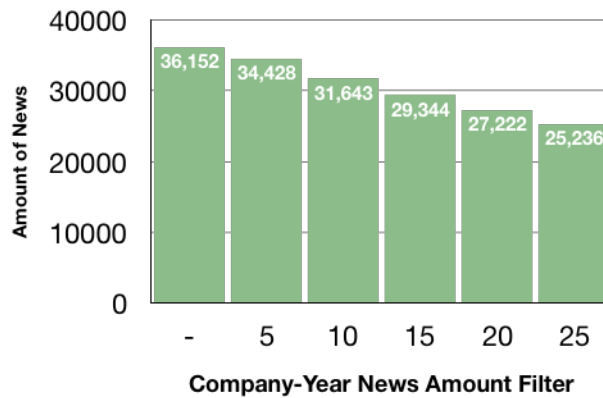


Figure 9: *Statistical Summary of Number of News Articles with Different Thresholds of Company-year News Amount Filter*

We observe that after setting the threshold of each company-year news amount, the number of the company-years has dramatically dropped 60% from 2,050 to 796 when the threshold is set from 0 to 10. Yet, the total amount of news merely has a slight decrease of 12% from 36,152 to 31,643. It suggests that the news articles we have gathered are all from the same few companies. We have made various filter hypotheses and found better model performances when the news amount threshold is set to 20.

Because of the limited data size and the data imbalance for our model training, we group the 7 MSCI ESG Ratings (Indexes) into 3 ESG classes for our classification task: We form ratings of $\{AAA, AA, A\}$ into the “High” class, $\{BBB\}$ into the “Medium” class and $\{BB, B, CCC\}$ into the “Low” class. The distributions of MSCI ESG indexes, ESG classes, and the ESG news articles of each ESG class are shown in Figure 10, Figure 11, and Figure 12, respectively.

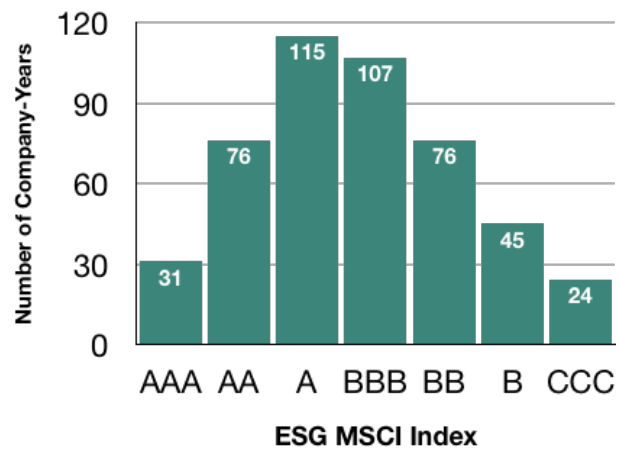


Figure 10: *Distribution of Number of Company-years by ESG MSCI Indexes (with Company-year News Amount Threshold Set to 20)*

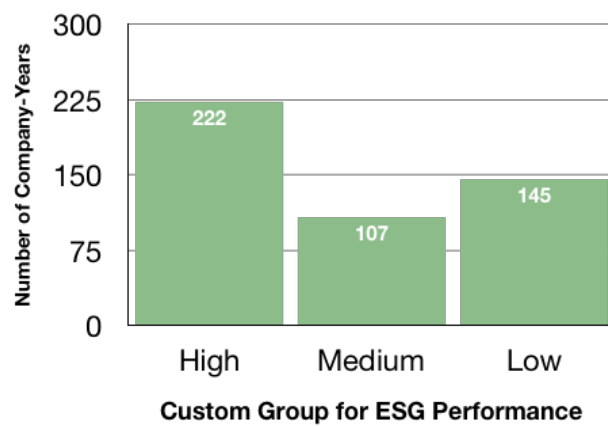


Figure 11: *Distribution of Number of Company-years by Thee ESG Classes (with Company-year News Amount Threshold Set to 20)*

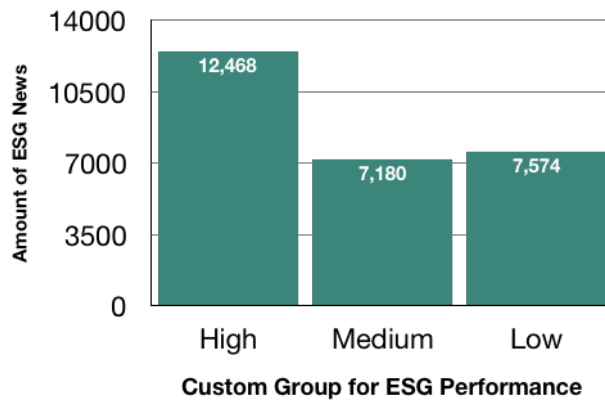


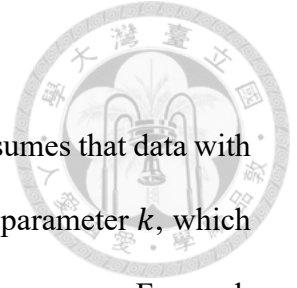
Figure 12: *Distribution of ESG News Amount by Three ESG Classes (with Company-year News Amount Threshold Set to 20)*

4.2 Classification Algorithms

In the proposed vector space model and the news embedding via BERT encoder model, we use the following classification algorithms to train classification models.

- k nearest neighbors
- random forest
- XGBoost
- backpropagation neural network

While in BERT for classification and LSTM with embedding layer, we append fully connected layers on top of the model. The detailed hyperparameters are described in Section 4.3.2.



4.2.1 k Nearest Neighbors

k NN is a supervised and non-parametric learning algorithm. It assumes that data with similar output are nearby. In this algorithm, we first need to decide the parameter k , which is referred to the number of nearest neighbors to include in the voting process. For each unlabeled data point, k NN will vote the classification outcome by the k nearest neighbors. Since the model does not make any assumption on the underlying data distribution and uses the voting mechanism, it is robust to noisy data.

4.2.2 Random Forest

Random forest is an algorithm consisting of various decision trees, so-called a bagging algorithm. Individual trees in the random forest are relatively uncorrelated and will perform their classification tasks under different features and circumstances. The class with the most votes from the various decision trees will become the model prediction outcome.

4.2.3 XGBoost

XGBoost is short for “eXtreme Gradient Boosting,” which is also an algorithm consisting of various decision trees. Different from random forest, XGBoost will further calculate the loss function using gradient boosting. In this way, XGBoost can reduce the bias of individual models made to improve performance. Yet, because the model needs to be fine-tuned and is easy to overfit the data. Sometimes, it is not guaranteed to outperform the bagging (random forest) algorithm.



4.2.4 Backpropagation Neural Network

In this part, our backpropagation neural network only performs the target classification task with multiple fully connected layers. Fully connected layers are those layers that connect every neuron in one layer to every neuron of the next layer. Our backpropagation neural network classifier consists of two to three fully connected layers, which are fed with the vector or representation of a company-year sample and make a final prediction for the focal company-year.

4.3 Experiment Setting

4.3.1 Evaluation Metrics

First, accuracy as well as macro F1 and weighted F1 are used to assess a model's classification effectiveness. Then we will focus on the model with the best result and further explore the performance of each class using precision, recall, and F1 score. The evaluation metrics employed in this study are shown in Equation 6.

$$accuracy(\mathbf{y}, \mathbf{p}) = \frac{1}{N} \sum_{i=1}^N 1(p_i = y_i) \quad (6)$$

$$precision_c = \frac{tp_c}{tp_c + fp_c}$$

$$recall_c = \frac{tp_c}{tp_c + fn_c}$$

$$F1_c = 2 \times \frac{precision_c \times recall_c}{precision_c + recall_c}$$

$$macro\ F1 = \frac{1}{|C|} \sum_{c \in C} F1_c$$

$$weighted\ F1 = \frac{N_c}{N} \sum_{c \in C} F1_c$$



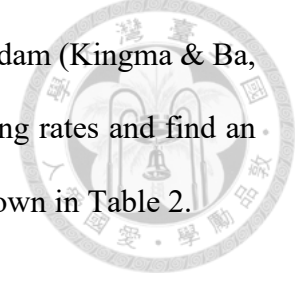
where N is the total number of testing samples, $c \in \text{ESG classes}$, N_c is the number of testing samples that belong to the ESG class c .

4.3.2 Training Strategies and Hyperparameters Setting

We first discuss the hyperparameters of the four different classification algorithms mentioned in Section 4.2, which are used by the vector space model and the news embedding via BERT encoder model.

- In the k NN classifier, we set the parameter k as 5.
- In random forest classifier, we set the number of trees to 100.
- In XGBoost classifier, we set the learning rate to 0.1, the number of trees to 100, and the maximum depth of a tree to 3.
- In the backpropagation neural network classifier, there exist slight differences between the two models. For the vector space model, we build the classifier with three fully connected layers, whose hidden units are 128, 32, and 3 respectively. For the news embedding via BERT encoder model, the classifier consists of two fully connected layers, whose hidden units are 128 and 3. We adopt an early stopping strategy during our training process with a patience window set to 20 instead of setting a fixed number of training epochs. We then adopt the cross-

entropy (shown in Equation 7) as the loss function and use Adam (Kingma & Ba, 2014) as the optimizer. We experiment with different learning rates and find an optimal learning rate of $1e - 4$. The hyperparameters are shown in Table 2.



$$CE(y, p) = - \sum_{c=1}^c y_c \log p_c \quad (7)$$

where y_c = ground truth and p_c = predicted probability of class c .

Table 2: *Hyperparameters for the Backpropagation Neural Network*

Learning rate	Patience window	Adam parameters
$1e - 4$	20	$\beta_1: 0.9, \beta_2: 0.999$

In the BERT for classification model, we adopt the ESG-BERT model and append a fully connected layer with input/output dimensions (768, 3) on top of ESG-BERT. Due to the limitation of our GPUs, we can only set our batch size as 8. We adopt the cross-entropy as the loss function and use Adam as the optimizer. The hyperparameters for the BERT for classification model are shown in Table 3.

Table 3: *Hyperparameters of the BERT for Classification*

Batch size	Learning rate	Epochs	Adam parameters
8	$5e - 5$	3	$\beta_1: 0.9, \beta_2: 0.999$

For the LSTM with Embedding Layer model, we build our model with one embedding layer, two bidirectional LSTM layers, and two fully connected layers. We adopt an early stopping strategy, the cross-entropy as the loss function, and use Adam as the optimizer. The hyperparameters for the LSTM with Embedding Layer model are shown in Table 4.

Table 4: *Hyperparameters for the LSTM with Embedding Layer Model*

Embedding dimensions	Learning rate	Patience	Adam parameters	Dropout
128	$2e - 4$	20	$\beta_1: 0.9, \beta_2: 0.999$	0.1
Bi-LSTM1 input/output size	Bi-LSTM2 input/output size	FC layer1 input/output size	FC layer2 input/output size	
(128, 256)	(256, 128)	(139*, 32)	(32, 3)	

*: The first fully connected layer will receive from the Bi-LSTM2 layer with 128 dimensions and company sectors with 11 dimensions.

4.4 Experimental Results and Discussions

In this section, all models are trained with the news amount threshold set to 20. Therefore, 474 company-years remain in our dataset. Each experimental result is derived from 3 times of 10-fold cross-validation with different random seeds.

4.4.1 Vector Space Model

The evaluation result of the vector space model is shown in Table 5. We observe that the XGBoost classifier reaches the highest accuracy when the number of SVD dimensions is reduced to 200, and the backpropagation neural network (NN) classifier reaches the highest

macro F1 score when the number of SVD dimensions is reduced to 300. Besides, we can see that adopting dimension reduction also improves the classification effectiveness when random forest (RF) classifier or the k NN classifier is employed.

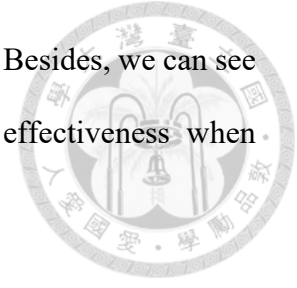


Table 5: *Evaluation Result of the Vector Space Model*

Algorithm \ SVD Dim.	Score: (Accuracy / Macro F1)			
	k NN	RF	NN	XGBoost
None	0.4437 / 0.2881	0.4388 / 0.2775	- / -	- / -
100	0.4810 / 0.3098	0.4782 / 0.3071	0.4684 / 0.4050	0.4620 / 0.3821
200	0.4761 / 0.2883	0.4859 / 0.2912	0.4522 / 0.3932	0.4873 / 0.3934
300	0.4698 / 0.2625	0.4782 / 0.2796	0.4754 / 0.4135	0.4873 / 0.3933
400	0.4810 / 0.2692	0.4845 / 0.2727	- / -	0.4747 / 0.3674

Please note that boldface denotes the highest score across all of the models.

Because of the giant size of news vectors (>20k dimension), we are not able to conduct experiments without dimension reduction when using NN and XGBoost classifiers (i.e., our GPU maximum limit is 11,016 MiB, which is not enough for storing the original news vectors). Moreover, while the news amount threshold is set to 20, there are only 474 company-years remained in our dataset. Since the NN classifier needs to split data into training, testing, and validation sets, the data size is not big enough for the classifier to do so when the number of dimensions is reduced to 400 by SVD. Across all the models, we observe that the highest classification effectiveness is attained when the number of SVD dimensions is set to 200. Accordingly, we further explore the detailed evaluation results when the number of SVD dimensions is set to 200 and show the result in Table 6.

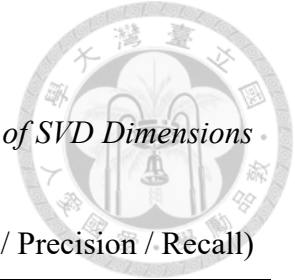


Table 6: *Evaluation Result of the Vector Space Model with the Number of SVD Dimensions*
Set to 200

Score: (F1 / Precision / Recall)

Algorithm \ Class	kNN	RF	NN	XGBoost
Low	0.2195 / 0.4769 / 0.1425	0.2152 / 0.5577 / 0.1333	0.4048 / 0.4253 / 0.3862	0.4444 / 0.4627 / 0.4276
Medium	0.0116 / 0.0833 / 0.0062	0.0178 / 0.1765 / 0.0093	0.1616 / 0.1645 / 0.1589	0.1216 / 0.2195 / 0.0841
High	0.6339 / 0.4834 / 0.9204	0.6406 / 0.4842 / 0.9459	0.6132 / 0.5914 / 0.6366	0.6142 / 0.5351 / 0.7207
Macro Avg.	0.2883 / 0.3479 / 0.3564	0.2912 / 0.4061 / 0.3629	0.3932 / 0.3937 / 0.3939	0.3934 / 0.4058 / 0.4108
Weighted Avg.	0.3667 / 0.3911 / 0.4761	0.3699 / 0.4372 / 0.4859	0.4475 / 0.4442 / 0.4522	0.4511 / 0.4417 / 0.4873

Please note that boldface denotes the highest score across all of the models.

We observe that the XGBoost classifier outperforms all the other classifiers in macro F1 and weighted F1 scores. Due to the class imbalance problem, each score in weighted average is 5%-8% higher than the macro average. In addition, the recall of the high class in the RF classifier has reached 94.59% yet only with the precision of 48.42%, which implies that RF is tempt to guess the outcome of the high class. We also find that models face difficulties in classifying the medium class with the highest F1 score of only 16.16% when using the NN classifier.

4.4.2 News Embedding via BERT Encoder Model

The evaluation result of the news embedding via BERT encoder model is shown in Table 7. We observe that the models using ESG-BERT encoder completely beat the models using the uncased-BERT encoder with an over 10% improvement in k NN, RF, and XGBoost classifiers, emphasizing the power of further pre-training. Besides, we also find that adopting the ESG weighted average (ESG_WA) scheme performs better in most of the cases. This demonstrates that our ESG weighted average scheme is effective.

Table 7: *Evaluation Result of the News Embedding via BERT Encoder Model*

Score: (Accuracy / Macro F1)				
Algorithm \ Average	k NN	RF	NN	XGBoost
uncased-BERT				
NA	0.4444 / 0.2578	0.4437 / 0.2617	0.4965 / 0.3728	0.4395 / 0.3256
ESG_WA	0.4487 / 0.2537	0.4444 / 0.2540	0.5000 / 0.3824	0.4444 / 0.3388
ESG-BERT				
NA	0.5492 / 0.4618	0.5457 / 0.4587	0.5309 / 0.4354	0.5619 / 0.4985
ESG_WA	0.5513 / 0.4622	0.5422 / 0.4506	0.5394 / 0.4206	0.5724 / 0.5064

Please note that boldface denotes the highest score across all of the models.

We further explore the model using the ESG-BERT encoder with the ESG weighted average scheme. The evaluation result is shown in Table 8. In this specific setting, we observe that XGBoost outperforms other classifiers. Besides, we can see that the F1 score in the medium class has a significant improvement compared to the vector space model.

Table 8: *Evaluation Result of the News Embedding via BERT Encoder Model using ESG-BERT with ESG Weighted Average Scheme*

Score: (F1 / Precision / Recall)

Algorithm \ Class	kNN	RF	NN	XGBoost
Low	0.4378 / 0.5405 / 0.3678	0.4163 / 0.5100 / 0.3517	0.4695 / 0.4612 / 0.4782	0.4975 / 0.5358 / 0.4644
Medium	0.2729 / 0.4841 / 0.1900	0.2599 / 0.4436 / 0.1838	0.1105 / 0.3559 / 0.0654	0.3287 / 0.4511 / 0.2586
High	0.6759 / 0.5630 / 0.8453	0.6755 / 0.5652 / 0.8393	0.6819 / 0.5899 / 0.8078	0.6929 / 0.6144 / 0.7943
Macro Avg.	0.4622 / 0.5292 / 0.4677	0.4506 / 0.5063 / 0.4583	0.4206 / 0.4690 / 0.4505	0.5064 / 0.5338 / 0.5057
Weighted Avg.	0.5121 / 0.5383 / 0.5513	0.5024 / 0.5209 / 0.5422	0.4879 / 0.4977 / 0.5394	0.5509 / 0.5535 / 0.5724

Please note that boldface denotes the highest score across all of the models.

4.4.3 BERT for Classification

In the proposed BERT for classification model, we can see that ESG-BERT is always guessing the same outcome of the high class and does not learn anything from our data, as Table 9 illustrates. Due to the constraint of only allowing a small number of news articles (i.e., 8) to be fed into the model, it is difficult for ESG-BERT to learn from them. Besides, we cannot ensure the news articles we randomly choose for each batch are representative enough to be the news data for the company-year. Moreover, the data size (i.e., the number of company-years) is only 474, which might be too small for ESG-BERT to learn. As we know the classes are imbalanced (shown in Figure 11), the model is tempted to predict the class with the greatest number of company-years in the training dataset.

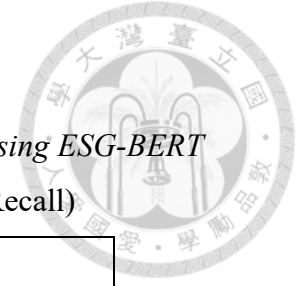


Table 9: *Evaluation Result of the BERT for Classification Model Using ESG-BERT*

Score: (F1 / Precision / Recall)

Class \ Model	ESG-BERT
Low	- / - / -
Medium	- / - / -
High	0.6379 / 0.4684 / 1.0000
Macro Avg.	- / - / -
Weighted Avg.	- / - / -

4.4.4 LSTM with Embedding Layer

We also find the model of LSTM with embedding layer cannot learn well, as Table 10 shows. Even for the high class, the recall rate is only 57.29%, which is much lower than the models reported in the previous subsections. During the training stage, we find that this particular model is very easy to overfit the training data. We attribute this problem to the limited training data size. For using a dataset with only 474 company-years, it is difficult to train an embedding layer from the scratch.

Table 10: Evaluation Result of LSTM with Embedding Layer

Score: (F1 / Precision / Recall)

Class \ Model	LSTM with Embedding Layer
Low	0.3704 / 0.3467 / 0.3977
Medium	0.1896 / 0.2350 / 0.1589
High	0.5729 / 0.5567 / 0.5901
Macro Avg.	0.3776 / 0.3795 / 0.3822
Weighted Avg.	0.4244 / 0.4198 / 0.4339

4.4.5 Model Comparison and Discussions

In Table 11, we observe that using the news embedding via BERT encoder model has reached the best performance with the macro F1 score of 50.64% and weighted F1 score of 55.09%. We attribute this to the BERT encoder since BERT can learn the semantic information of news articles while the vector space model can only learn the lexicon-based information. Furthermore, we find that it is difficult to train a deep learning-based model from the scratch when the training data size is very limited. Yet, despite the constraint of our GPU resources, we believe if the batch size can be larger, ESG-BERT might achieve a better performance.

Table 11: Evaluation Result of Different Models with the Optimal Performance

Score: (F1 / Precision / Recall)

Model \ Class	Vector Space Model SVD: 200 Algorithm: XGBoost	News Embedding via BERT Encoder Model BERT: ESG-BERT AVG.: ESG_WA Algorithm: XGBoost	BERT for Classification	LSTM with Embedding Layer
Low	0.4444 / 0.4627 / 0.4276	0.4975 / 0.5358 / 0.4644	- / - / -	0.3704 / 0.3467 / 0.3977
Medium	0.1216 / 0.2195 / 0.0841	0.3287 / 0.4511 / 0.2586	- / - / -	0.1896 / 0.2350 / 0.1589
High	0.6142 / 0.5351 / 0.7207	0.6929 / 0.6144 / 0.7943	0.6379 / 0.4684 / 1.0000	0.5729 / 0.5567 / 0.5901
Macro Avg.	0.3934 / 0.4058 / 0.4108	0.5064 / 0.5338 / 0.5057	- / - / -	0.3776 / 0.3795 / 0.3822
Weighted Avg.	0.4511 / 0.4417 / 0.4873	0.5509 / 0.5535 / 0.5724	- / - / -	0.4244 / 0.4198 / 0.4339

Please note that boldface denotes the highest score across all of the models.

4.5 Additional Experiments

We also conduct additional experiments over some factors of models. First, we demonstrate the impact of setting different news amount filter thresholds. The evaluation result is shown in Table 12. We observe that when the news amount threshold is set to 20, all models will have their best performance. Thus, in the research, we choose 20 as our threshold. We can also see the news embedding via BERT encoder model always attains the best performance across all models at any news amount filter threshold investigated.

Table 12: *Evaluation Result of Various News Amount Filter Thresholds by Different Optimized Models*

Score: (Accuracy / Macro F1)

Model \ News Threshold	Vector Space Model SVD: 200 Algorithm: XGBoost	News Embedding via BERT Encoder Model BERT: ESG-BERT AVG.: ESG_WA Algorithm: XGBoost	LSTM with Embedding Layer
10	0.4510 / 0.3262	0.5607 / 0.4731	0.4037 / 0.3415
15	0.4317 / 0.3266	0.5444 / 0.4691	0.4100 / 0.3715
20	0.4873 / 0.3934	0.5724 / 0.5064	0.4339 / 0.3776
25	0.4308 / 0.3218	0.5527 / 0.4909	0.4151 / 0.3393

Please note that boldface denotes the highest score across all of the models.

Second, with regard to the imbalanced classes, we have re-grouped our MSCI Ratings into 2 ESG classes: we merge the “low” class and the “medium” class into one class while the “high” class stands for its own class. In this way, the data will become more balanced. We use the news embedding via BERT encoder model, which has reached the best performance in the 3-class classification task. As Table 13 illustrates, we observe that when we group the ESG Ratings into 2 classes, the model has a satisfactory performance of 68.60% in macro F1 score. Yet, we still expect our model can classify 3 classes more effectively instead of 2, predicting one firm’s ESG performance more precisely.

Table 13: *Evaluation Result of 2 vs 3 ESG Classes Using the News Embedding via BERT*
Encoder Model (News Filter Threshold = 20, BERT: ESG-BERT; Average: ESG_WA)

Score: (F1 / Precision / Recall)

Model \ Class	2 ESG Classes	3 ESG Classes
Low	0.7174 / 0.6943 / 0.7421	0.4975 / 0.5358 / 0.4644
Medium		0.3287 / 0.4511 / 0.2586
High	0.6547 / 0.6824 / 0.6291	0.6929 / 0.6144 / 0.7943
Macro Avg.	0.6860 / 0.6884 / 0.6856	0.5064 / 0.5338 / 0.5057
Weighted Avg.	0.6880 / 0.6887 / 0.6892	0.5509 / 0.5535 / 0.5724

Please note that boldface denotes the highest score across all of the models.

Chapter 5

Conclusion



ESG is commonly agreed to be a key agenda in every industry nowadays. Companies will strive to achieve their sustainability goals while agencies and investors will undertake various measures to evaluate the companies' ESG performance. In the research, we use the machine learning approach to propose several automatic methods, which can dynamically predict the ESG performance of a company when given a time period using publicly available information (news articles in this study). Also, by predicting ESG classes automatically, we believe we can save time on evaluating one firm's ESG performance and solve the problem of companies do not have any ratings by agencies.

According to our evaluation results, we find out the news embedding via BERT encoder model achieves the best classification effectiveness among all models. We conclude that because of the architecture and pre-training steps of BERT, the BERT encoder is able to catch semantic information from news data. Yet, due to the limitation of the data size and GPU constraints, deep learning-based models as the BERT for classification model and the LSTM with embedding layer model cannot learn well when training with limited training data. In addition, for the deep learning-based models, we can see that ESG-BERT outperforms uncased-BERT when the news embedding via BERT encoder model is

employed, which shows the magic of further pre-training BERT on domain-specific corpora. Besides, we propose an attention-like ESG weighted average method, which indeed improves the model performance.



Since ESG is a concept becoming important recently, there is still not much news talking about a company ESG related issues. As our task is focusing on “automatic” ESG performance classification via text-mining on ESG news articles, we believe the model performance can be improved when the news data size getting bigger in the future. Besides, future research can broaden datasets by targeting at or involving social media posts (e.g., Twitter, Reddit) which are related to companies’ ESG events.

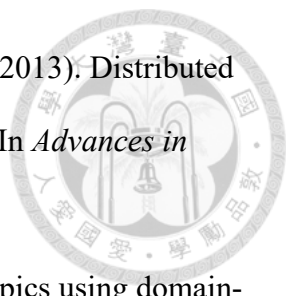
Another issue that we have not addressed in the current study is the multiple aspects of ESG. ESG contains three main subjects (i.e., environmental, social, and governance), while there are a variety of sub-concepts inside each of them. However, our model does not target at distinguishing the performance of a focal company in each of these subjects, but can only produce an overall ESG performance class for the focal company. Therefore, in the future, we look forward to extend our proposed methods by decomposing ESG into various subjects. With this, we can turn the classification task into a multi-classifier task with different sub-models to construct a more exquisite ESG classification model.

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Appendix A — *E-Description Document*



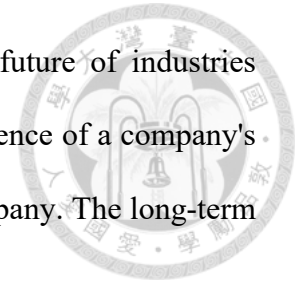
Threat of climate change and the depletion of resources has grown, so investors may choose to factor sustainability issues into their investment choices. The issues often represent externalities, such as influences on the functioning and revenues of the company that are not exclusively affected by market mechanisms. As with all areas of ESG, the breadth of possible concerns is vast (e.g., greenhouse gas emissions, biodiversity, waste management, water management) but some of the chief areas are listed below:

Climate crisis

The body of research providing evidence of global trends in climate change has led investors—pension funds, holders of insurance reserves—to begin to screen investments in terms of their impact on the perceived factors of climate change. Fossil fuel reliant industries are less attractive. In the UK, investment policies were particularly affected by the conclusions of the Stern Review in 2006, a report commissioned by the British government to provide an economic analysis of the issues associated with climate change. Its conclusions pointed towards the necessity of including considerations of climate change and environmental issues in all financial calculations and that the benefits of early action on climate change would outweigh its costs.

Sustainability

In every area of the debate from the depletion of resources to the future of industries dependent upon diminishing raw materials the question of the obsolescence of a company's product or service is becoming central to the value ascribed to that company. The long-term view is becoming prevalent amongst investors.



Appendix B — *S-Description Document*



Diversity

The level of diversity as well as inclusion in a company's recruitment and people management policies is becoming a key concern to investors. There is a growing perception that the broader the pool of talent open to an employer the greater the chance of finding the optimum person for the job. Innovation and agility are seen as the great benefits of diversity, and there is an increasing awareness of what has come to be known as 'the power of difference'.

Human rights

In 2006 the US Courts of Appeals ruled that there was a case to answer bringing the area of a company's social responsibilities squarely into the financial arena. This area of concern is widening to include such considerations as the impact on local communities, the health and welfare of employees and a more thorough examination of a company's supply chain.

Consumer protection

Until fairly recently, caveat emptor ("buyer beware") was the governing principle of commerce and trading. In recent times however there has been an increased assumption that the consumer has a right to a degree of protection and the vast growth in damages litigation has meant that consumer protection is a central consideration for those seeking to limit a company's risk and those examining a company's credentials with an eye to investing. The

collapse of the US subprime mortgage market initiated a growing movement against predatory lending has also become an important area of concern.



Animal welfare

From the testing of products on animals to the welfare of animals bred for the food market, concern about the welfare of animals is a large consideration for those investors seeking a thorough understanding of the company or industry being analyzed.

Appendix C — *G-Description Document*



Corporate governance covers the area of investigation into the rights and responsibilities of the management of a company—its board, shareholders and the various stakeholders in that company.

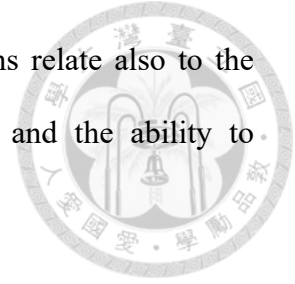
Management structure

The system of internal procedures and controls that makes up the management structure of a company is in the valuation of that company's equity.[16] Attention has been focused in recent years on the balance of power between the CEO and the Board of Directors and specifically the differences between the European model and the US model—in the US studies have found that 80% of companies have a CEO who is also the Chairman of the Board, in the UK and the European model it was found that 90% of the largest companies split the roles of CEO and Chairman.

Employee relations

From diversity to the establishment of corporate behaviors and values, the role that improving employee relations plays in assessing the value of a company is proving increasingly central. In the United States Moskowitz's list of the Fortune 100 Best Companies to Work For has become not only an important tool for employees but companies are beginning to compete keenly for a place on the list, as not only does it help to recruit the best workforce, it appears

to have a noticeable impact on company values.[8] Employee relations relate also to the representation of co-workers in the decision-making of companies, and the ability to participate in a union.



Executive compensation

Companies are now being asked to list the percentage levels of bonus payments and the levels of remuneration of the highest paid executives are coming under close scrutiny from stock holders and equity investors alike.

Employee compensation

Besides executive compensation, equitable pay of other employees is a consideration in the governance of an organization. This includes pay equity for employees of all genders. Pay equity audits and the results of those audits may be required by various regulations and, in some cases, made available to the public for review.